On conditional edge-fault-tolerant cycles embedding of hypercubes *

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Abstract

The hypercube has been one of the most popular interconnection networks for parallel computer/communication system. In this papers, we assume that each node is incident with at least two fault-free links. Under this assumption, we show that every fault-free edge lies on a fault-free cycle of every even length from 6 to 2^n inclusive even if it has up to $2n-5$ link faults. The result is optimal with respect to the number of link faults tolerated.
Keywords: interconnection network, hypercube, Hamiltonian, conditional fault-tolerant, bipancyclic.

1 Introduction

An interconnection network such as hypercube is conveniently represented with an undirected graph. The vertices (edges) of the graph denote the nodes (links) of the network. Throughout this paper, vertex and node, edge and link, graph and network are used interchangeably. For the graph definition and notation we follow [5]. $G=(V,E)$ is a graph if V is a finite set and E is a subset of $\{(a,b) \mid (a,b)$ is an unordered pair of $V\}$. We say that V is the *vertex set* and E is the *edge set*. A graph $G=(V_0 \cup V_1, E)$ is *bipartite* if $V(G)$ is the union of two disjoint sets V_0 and V_1 such that every edge joins V_1 with V_2. Two vertices a and b are *adjacent* if $(a,b) \in E$. Let F be a subset of $E(G)$. We use $G-F$ to denote the graph with vertex set $V(G)$ and edge set $E(G)-F$. A path is a sequence of adjacent vertices, written as $\langle v_0, v_1, v_2, \ldots, v_m \rangle$, in which all the vertices $v_0, v_1, \ldots, v_m$ are distinct except possibly $v_0 = v_m$. We also write the path $\langle v_0, P[v_0, v_m], v_m \rangle$, where $P[v_0, v_m] = \langle v_0, v_1 \ldots, v_m \rangle$ as well as v_0 and v_m are two *end-vertices* of $P[v_0, v_m]$. Two paths are vertex-disjoint (also call disjoint) if and only if they do not have any vertices in common. The *length* of a path $P[v_0, v_m]$ denoted by $l(P[v_0, v_m])$ is the number of edges in $P[v_0, v_m]$. A path $P[u,v]$ is called k-path joining u and v if $l(P[u,v]) = k$. Sometimes, we also use $P_k[u,v]$ to denote a k-path that joins u and v. Let u and v be two vertices of G. The *distance* between u and v denoted by $d_G(u,v)$ is the length of the shortest path of G joining u and v. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A cycle C is called k-cycle if $l(C) = k$. A k-cycle is *odd* or *even* depending on whether k is odd or even. A *Hamiltonian cycle* is a cycle of length $|V(G)|$; A *Hamiltonian path* is a path of length $|V(G)|-1$.

Let $u = u_{n-1}u_{n-2}\ldots u_1u_0$ be an n-bit binary strings. For $0 \le k < n$, we use u^k to denote the binary string $v_{n-1}v_{n-2}\ldots v_1v_0$ such that $v_k = 1-u_k$ and $u_i = v_i$ if $i \ne k$. Let $u = u_{n-1}u_{n-2}\ldots u_1u_0$ and $v = v_{n-1}v_{n-2}\ldots v_1v_0$ be two n-bit binary strings. The *Hamming distance* $h(u,v)$ between two vertices u and v is the number of different bits in the corresponding strings of both vertices. The *n-dimensional hypercube*, denoted by Q_n, consists of all n-bit binary strings as its vertices and two vertices u and v are adjacent if and only if $h(u,v) = 1$. An edge (u,v) in $E(Q_n)$ is of *dimension* i if $u = v^i$. It is known that $d_{Q_n}(u,v) = h(u,v)$. Subsequently, some properties of the hypercube are derived, which are useful in order to construct a fault-free cycle in a faulty hypercube.

Lemma 1 *Let e be any edge of Q_n for $n \ge 2$.*

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Then, there are $n-1$ cycles of length four that contain e in common.

Proof. Let $e=(u,v)$ and $v=u^i$ for some $0\le i\le n-1$, i.e., e is an edge of dimension i. Since $h(u,v)=1$, $h(u^j,v^j)=1$ for all $0\le j\le n-1$. Thus, a 4-cycle containing the edge (u,v) is formed by $\langle u,u^j,v^j,v,u\rangle$ for $0\le j\le n-1$ and $j\ne i$. For any vertex x in Q_n, $x^j\ne x^k$ if and only if $j\ne k$. One can observe that the value of j has $n-1$ possible values if the value of i is fixed. Therefore, there are $n-1$ cycles of length four which contain (u,v) in common. □

Lemma 2 *[6] For any subset F of $E(Q_n)$ with $|F|\le n-2$, every edge of Q_n-F lies on a cycle of every even length from 4 to 2^n inclusive for $n\ge 3$.*

Lemma 3 *[9] For any subset F of $V(Q_n)$ with $|F|\le n-2$, every edge of Q_n-F lies on a cycle of every even length from 4 to $2^n-2|F|$ inclusive for $n\ge 3$.*

Lemma 4 *[10] Any two edges of Q_n ($n\ge 2$) are included in a Hamiltonian cycle.*

Lemma 5 *[10] For any subset F of $E(Q_n)$ with $|F|\le n-1$, every edge of Q_n-F lies on a cycle of every even length from 6 to 2^n inclusive provided $n\ge 4$ and all edges in F are not incident with the same vertex.*

Lemma 6 *For $n\ge 2$, let (u,v) and (w,z) be two disjoint edges in Q_n. Then, Q_n can be partitioned into two $(n-1)$-cubes such that one contains (u,v) and the other contains (w,z).*

Proof. The lemma is true when $n=2$. Let (u,v) and (w,z) be two disjoint edges, and $v=u^i$ and $z=w^k$. Hence u, v, w, and z are four distinct vertices. Without loss of generality, we may assume that $u=00\ldots0$ and $w=w_{n-1}w_{n-2}\ldots w_{k+1}0w_{k-1}\ldots w_0$. Since $n\ge 3$ as well as u, v, and w are distinct, there exists $j\ne i$ and $j\ne k$ such that $w_j=1$. One may partition Q_n along dimension j into two $(n-1)$-cubes, Q^0_{n-1} and Q^1_{n-1}, such that Q^0_{n-1} contains u and v as well as Q^1_{n-1} contains w and z. □

Lemma 7 *For any two disjoint edges (u,v) and (w,z) in Q_n with $n\ge 2$, there exist two disjoint paths $P_r[u,v]$ and $P_s[w,z]$, in Q_n where $r=1, 3, 5, 7,\ldots, 2^{n-1}-1$ and $s=1, 3, 5, 7,\ldots, 2^{n-1}-1$.*

Proof. Let (u,v) and (w,z) be two disjoint edges in Q_n. By Lemma 6, Q_n can be partitioned along dimension j into two $(n-1)$-cubes, Q^0_{n-1} and Q^1_{n-1}, such that Q^0_{n-1} contains (u,v) and Q^1_{n-1} contains (w,z) for some $0\le j\le n-1$. By Lemma 2, there exist paths joining u and v (respectively, w and z) of lengths 1, 3, …, $2^{n-1}-1$ in Q^0_{n-1} (respectively, Q^1_{n-1}). □

Definition 1 *A vertex u is called* ***i*-rescuable** *if there are exactly i fault-free edges incident with u. A vertex u is called* **isolated** *if none of fault-free edge is incident with u.*

Let F be a subset of $E(Q_n)$ and u be a fault-free vertex in Q_n. We define $N(u)=\{i\mid (u,u^i)$ as being a fault-free edge for $0\le i\le n-1\}$ and $\overline{N}(u)=\{i\mid (u,u^i)\in F\}$. Note that $|N(u)|$ denotes the number of fault-free edges incident with u, and $|\overline{N}(u)|$ denotes the number of faulty edges incident with u. Obviously, $N(u)\cap\overline{N}(u)=\emptyset$ and $N(u)\cup\overline{N}(u)=\{0,1,2,\ldots,n-1\}$.

2 Partition rules

Throughout this section, F is a subset of $E(Q_n)$ and every vertex is incident with at least two fault-free edges. Also let $f_e=|F|$, i.e., f_e denotes the number of faulty edges of Q_n. Therefore, none of 1-rescuable vertex exists in Q_n-F, i.e., $|N(u)|\ge 2$ for all vertex u in Q_n-F. Suppose that $f_e\le 2n-5$. Let u and v are distinct 2-rescuable vertices of Q_n-F. Consequently, there are $n-2$ faulty edges to be incident with u and v, respectively. Using the simple arithmetic, Lemma 8 is easy obtained.

Lemma 8 *Let $n\ge 4$, $F\subset E(Q_n)$ with $f_e\le 2n-5$, and every vertex is incident with at least two fault-free edges. Then,* **(1)** *At most two vertices are 2-rescuable.* **(2)** *If u and v are distinct 2-rescuable vertices in Q_n, $(u,v)\in F$.*

We partition Q_n along dimension j according to the following rules in the proofs of Theorem 1.

Rule 1: Suppose that Q_n contains exactly two vertices, u and v, to be 2-rescuable. Thus, $(u,v)\in F$. Let $v=u^j$. One may partition Q_n along dimension j into two $(n-1)$-cubes, denoted by Q^0_{n-1} and Q^1_{n-1}. Consequently, (u,v) be a crossing edge and every vertex of Q^0_{n-1} (respectively, Q^1_{n-1}) is incident with at least two fault-free edges in Q^0_{n-1} (respectively, Q^1_{n-1}).

Rule 2: Suppose that Q_n contains exactly one vertex to be 2-rescuable. Let $|N(u)|=2$ and

$(u, u^j) \in F$. One may partition Q_n along dimension j into two $(n-1)$-cubes, denoted by Q^0_{n-1} and Q^1_{n-1}, such that one contains u^j and the other contains u. Therefore, every vertex of Q^0_{n-1} (respectively, Q^1_{n-1}) is incident with at least two fault-free edges in Q^0_{n-1} (respectively, Q^1_{n-1}).

Rule 3: Suppose that none of 2-rescuable vertex exists in Q_n. In this situation, every vertex of Q_n is incident with at least three fault-free edges. Let $(u, v) \in F$ be an edge of dimension j. We partition Q_n along dimension j into two $(n-1)$-cubes such that (u, v) is a crossing edge.

Lemma 9 *Let $n \geq 4$, $f_e \leq 2n-5$, and every vertex is incident with at least two fault-free edges. Assume that Q_n is partitioned into two $(n-1)$-cubes, Q^0_{n-1} and Q^1_{n-1}, by previous rules. Then, every vertex of Q^0_{n-1} (respectively, Q^1_{n-1}) is incident with at least two fault-free edges in Q^0_{n-1} (respectively, Q^1_{n-1}).*

Lemma 10 *Let $n \geq 4$, $f_e \leq 2n-5$, and every vertex is incident with at least two fault-free edges. Assume that Q_n is partitioned into two $(n-1)$-cubes, Q^0_{n-1} and Q^1_{n-1}, by previous rules. Then, every crossing edge (u, v) lies on a fault-free 6-cycle of $\langle u, w, x, y, z, v, u\rangle$ where $u, w, x \in V(Q^0_{n-1})$ and $y, z, v \in V(Q^1_{n-1})$.*

Proof. Let Q_n be partitioned along dimension j into two $(n-1)$-cubes, Q^0_{n-1} and Q^1_{n-1}. Let (u, v) be a fault-free crossing edge, and $u \in V(Q^0_{n-1})$ and $v \in V(Q^1_{n-1})$. Also let $|N(u) - \{j\}| = m_0$ and $|N(v) - \{j\}| = m_1$. Since every vertex in Q^0_{n-1} (respectively, Q^1_{n-1}) is incident with at least two fault-free edges in Q^0_{n-1} (respectively, Q^1_{n-1}), $m_0 \geq 2$ and $m_1 \geq 2$. Let $\mathcal{C}_6(u, v) = \{\langle u, w, x, y, z, v, u\rangle$ is a 6-cycle $|$ $u, w, x \in V(Q^0_{n-1})$ and $v, z, y \in V(Q^1_{n-1})\}$. The proof is divided into two parts: $|N(u) \cap N(v)| = 1$ and $|N(u) \cap N(v)| \geq 2$.

Suppose that $|N(u) \cap N(v)| = 1$, i.e., $N(u) \cap N(v) = \{j\}$. Let $i \in N(u) - \{j\}$ and $k \in N(v) - \{j\}$. Hence $i \neq k$. Let $w = u^i$, $x = w^k$, $z = v^k$, and $y = z^i$. One can observe that $\langle u, w, x, y, z, v, u\rangle$ is a 6-cycle in $\mathcal{C}_6(u, v)$. Therefore, there are at least $m_0 \times m_1$ 6-cycle in $\mathcal{C}_6(u, v)$. If none of this kind 6-cycle is fault-free, at least $2(n-1) - m_0 - m_1 + m_0 \times m_1$ edges must be fault. It means that $2(n-1) - m_0 - m_1 + m_0 \times m_1 \leq 2n-5$. This implies that $m_0 \times m_1 \leq m_0 + m_1 - 3$. It contradicts that $m_0 \times m_1 \geq m_0 + m_1$ for all $m_0 \geq 2$ and $m_1 \geq 2$. Therefore, a fault-free 6-cycle of $\langle u, w, x, y, z, v, u\rangle$ exists where $u, w, x \in V(Q^0_{n-1})$ and $y, z, v \in V(Q^1_{n-1})$. Similarly, it is not difficult to verify the case of that $|N(u) \cap N(v)| \geq 2$. The detail proof is omitted. □

3 Fault-tolerant cycle embedding

Let f_e be the number of faulty edges. We prove the main result of this section by an inductive schema.

Lemma 11 *Assume that $F \subseteq E(Q_n)$ and $F_a = F \cup \{a\}$ for any vertex a in Q_n. Then, every fault-free edge of $Q_n - F_a$ lies on a cycle of every even length from 4 to $2^n - 2$ provided $n \geq 3$ and $|F| \leq n-3$.*

Proof. Let $f_e = |F|$. The proof is induction on n. By Lemma 3, the lemma holds when $n = 3$. Assume that the lemma is true for every integer $3 \leq m < n$. We now consider $m = n$ as follows. If $f_e = 0$, one may choose any one edge (u, v) and imagine it is faulty. Thus, we only consider the case of that $1 \leq f_e \leq n-3$.

Let (u, v) be a faulty edge (or imagine a faulty edge) and $v = u^{(j)}$. One may partition Q_n along dimension j into two $(n-1)$-cubes, Q^0_{n-1} and Q^1_{n-1}, such that one contains the vertex u and the other contains the vertex v. Hence $f_c \geq 1$ if $f_e > 0$ and $f_c = 0$ if $f_e = 0$. Let f^i_e be the number of faulty edges in Q^i_{n-1} for $i = 0, 1$, and thus, $f_e \geq f^0_e + f^1_e$. Thus, $f^0_e \leq n-4$ and $f^1_e \leq n-4$. Without loss of generality, one may assume that a is in Q^0_{n-1}. Let (x, y) be any fault-free edge of Q_n. We will establish an even k-cycle containing (x, y) where $4 \leq k \leq 2^n - 2$. We check two possible distributions of the edge (x, y).

Case 1: $(x, y) \in E(Q^0_{n-1}) \cup E(Q^1_{n-1})$, i.e., (x, y) is in Q^0_{n-1} or in Q^1_{n-1}. We only consider that $(x, y) \in E(Q^0_{n-1})$ since the discussion of $(x, y) \in E(Q^1_{n-1})$ is the same.

Since $f^0_e \leq n-4$ and $a \in V(Q^0_{n-1})$, and by induction hypothesis, the desired cycle of every even length from 4 to $2^{n-1} - 2$ inclusive can be found in Q^0_{n-1}. Let C_h be a fault-free h-cycle containing the edge (x, y) in $Q^0_{n-1} - \{a\}$ where $h = 2^{n-1} - 2$. One can observe that there are at least $2^{n-2} - 1$ mutually disjoint edges in the cycle C_h such that each edge differs from (x, y). Since $n \geq 4$, $2^{n-2} > n-3$. Consequently, there exists at least one edge (c, d) in C_h such that (c, c^j), (d, d^j), and (c^j, d^j) are fault-free edges. Hence a fault-free path of $P_{2^{n-1}-3}[c,d]$ exists in Q^0_{n-1} such that it covers the edge (x,y).

Since $f^1_e \le n-4$ and by Lemma 2, there are even cycles with lengths from 4 to 2^{n-1} inclusive containing the edge (c^j,d^j) in Q^1_{n-1}. Hence a fault-free r-path of $P_r[c^j,d^j]$ in Q^1_{n-1} exists where $r = 1,3,\dots,2^{n-1}-1$. Merging the two paths $P_{2^{n-1}-3}[c,d]$ and $P_r[c^j,d^j]$ as well as the two edges (c,c^j) and (d,d^j), we can construct an even l-cycle which contains (x,y) where $l = 2^{n-1}+r-1$. Therefore, $2^{n-1} \le l \le 2^n-2$.

Case 2: $(x,y) \notin E(Q^0_{n-1}) \cup E(Q^1_{n-1})$, i.e., (x,y) is a crossing edge between Q^0_{n-1} and Q^1_{n-1}.

Obviously, (x,y) is an edge of dimension j and $y = x^j$. Without loss of generality, we may assume that $x \in V(Q^0_{n-1})$ and $y \in V(Q^1_{n-1})$. By Lemma 1, there are $n-1$ cycles with length four containing the edge (x,y) in common. Since $f_e \le n-3$, there exists an integer i such that $i \ne j$, $x^i \ne a$, and $\langle x,x^i,y^i,y,x\rangle$ is a fault-free 4-cycle. Since $f^0_e \le n-4$ and $f^1_e \le n-4$ as well as by induction hypothesis and by Lemma 2, there exists a fault-free s-path of $P_s[x,x^i]$ in Q^0_{n-1} and a fault-free t-path of $P_t[y,y^i]$ in Q^1_{n-1} where $P_s[x,x^i]$ does not pass a, $s = 1,3,5,\dots,2^{n-1}-3$, and $t = 1,3,5,\dots,2^{n-1}-1$. Merging these two paths $P_s[x,x^i]$ and $P_t[y,y^i]$ as well as the two edges (x,y) and (x^i,y^i), an even l-cycle is established that contains the edge (x,y) where $l = s+t+2$. Therefore, $4 \le l \le 2^n-2$. □

Lemma 12 *[1] For $n \ge 4$, Q_n contains a fault-free Hamiltonain cycle provided $f_e \le 2n-5$ and every vertex is incident with at least two fault-free edges.*

Lemma 13 *[8] Assume that $n \ge 4$, and x and y are in different partite sets. Then, Q_n contains a fault-free Hamiltonain path joining x and y provided $f_e \le 2n-5$ and every vertex is incident with at least two fault-free edges.*

Theorem 1 *For $n \ge 4$, every fault-free edge of Q_n lies on a fault-free cycle of every even length from 6 to 2^n inclusive provided $f_e \le 2n-5$ and every vertex is incident with at least two fault-free edges.*

Proof. By Lemma 5, the theorem holds if $f_e \le n-1$. Hence we only need to prove the case of that $n \le f_e \le 2n-5$. The following proof is induction on n. By Lemma 5, the theorem is true if $n = 4$. Assume that the result is true for $4 \le m < n$. We now consider that $m = n$. Applying partition rules in previous section, one may partition Q_n along some dimension j, $0 \le j \le n-1$, into two $(n-1)$-cubes, denoted by Q^0_{n-1} and Q^1_{n-1}, such that there is at least one faulty edge is crossing edge and every vertex of Q^0_{n-1} (respectively, Q^1_{n-1}) is incident with at least two fault-free edges in Q^0_{n-1} (respectively, Q^1_{n-1}). Let f^i_e be the number of faulty edges in Q^i_{n-1} for $i = 0,1$ and f_c be the number of faulty crossing edges . Hence $f_c \ge 1$ and $f^0_e + f^1_e \le 2n-6$. Without loss of generality, assuming that $f^0_e \ge f^1_e$. Thus, $n-3 \le f^0_e \le 2n-6$ and $0 \le f^1_e \le n-3$. Let $e = (u,v)$ be any fault-free edge of Q_n. We will establish a cycle of every even length from 6 to 2^n inclusive such that the edge e is included. We check two possible distributions of faulty edges as following two cases.

Case 1. $f^0_e = 2n-6$ and $f^1_e = 0$.

Subcase 1-1. $e \in E(Q^0_{n-1})$. Note that $e = (u,v)$. Since $n \ge 5$, $f^0_e \ge 4$. Let (a,b) be a faulty edge in Q^0_{n-1} such that (a,a^j) and (b,b^j) are two fault-free edges. By induction hypothesis (imagine (a,b) fault-free), (u,v) lies on an even k-cycle of C_k in Q^0_{n-1} where $k = 6,8,\dots,2^{n-1}$. One may choose an edge (x,y) in C_k satisfies that $(x,y) = (a,b)$ if C_k passes the edge (a,b) or any edge (x,y) except (u,v) in C_k such that (x,x^j) and (y,y^j) are fault-free if C_k does not contain the edge (a,b). Therefore, a fault-free s-path of $P_s[x,y]$ in Q^0_{n-1} exists where $P_s[x,y]$ passes the edge (u,v) and $s = 5,7,\dots,2^{n-1}-1$. Since Q^1_{n-1} is fault-free and by Lemma 2, a fault-free r-path of $P_r[x^j,y^j]$ in Q^1_{n-1} exists for $r = 1,3,\dots,2^{n-1}-1$. The desired even cycle of every length from 8 to 2^n inclusive can be constructed by the combination of $P_s[x,y]$ and $P_r[x^j,y^j]$ as well as (x,x^j) and (y,y^j). The desired 6-cycle is easy to establish. Here the detail of construction is omitted.

Subcase 1-2. $e \in E(Q^1_{n-1})$. Since Q^1_{n-1} is fault-free, the desired even cycle of every length from 4 to 2^{n-1} is obtained by Lemma 2. Let (a,b) be a faulty edge in Q^0_{n-1} such that $\{a^j,b^j\} \cap \{u,v\} = \emptyset$, and (a,a^j) and (b,b^j) are two fault-free edges. By Lemma 4, let $P_{2^{n-1}-1}[a^j,b^j]$ be a Hamiltonian path of Q^1_{n-1} which contains the edge (u,v). By induction hypothesis (imagine (a,b) fault-free), (a,b) lies on a cycle of every even length from 6 to 2^{n-1} inclusive. This implies that a fault-free k-path of $P_k[a,b]$ in Q^0_{n-1} exists for $k = 5,7,\dots,2^{n-1}-1$. Combining $P_{2^{n-1}-1}[a^j,b^j]$ and $P_k[a,b]$ as well as (a,a^j) and (b,b^j), the desired cycle of every length from $2^{n-1}+6$ to 2^n inclusive.

Hereafter, two cycles of lengths $2^{n-1}+2$ and $2^{n-1}+4$ will be established. Let C_k be a Hamiltonian cycle of Q^1_{n-1} which contains the edge (u,v).

Hence there are 2^{n-2} mutually disjoint edges excluding (u, v). Since $2^{n-2} - (2n-5) > 2$ for $n \geq 5$. This implies that there exist two disjoint edges (x, y) and (w, z) in C_k such that (x, x^j), (y, y^j), (x^j, y^j), (w, w^j), (z, z^j), and (w^j, z^j) are fault-free. The desired fault-free cycle of length $2^{n-1} + 2$ can be constructed by the aid of edges (x, x^j), (y, y^j), (x, y), and the cycle C_k. Similarly, the desired fault-free cycle of length $2^{n-1} + 4$ can be constructed by the aid of edges (x, x^j), (y, y^j), (x, y), (w, w^j), (z, z^j), (w, z), and the cycle C_k.

Subcase 1-3. e is a crossing edge. Without loss of generality, we may assume that $u \in V(Q^0_{n-1})$ and $v \in V(Q^1_{n-1})$. Let $w(= u^i)$ and $z(= u^k)$ are distinct vertices in Q^0_{n-1} such that (u, w) and (u, z) are fault-free. This implies that $i \neq k \neq j$. Since $f^1_e = 0$ and $f_c = 1$, either (w, w^j) or (z, z^j) is fault-free. Without loss of generality, one may assume that (w, w^j) is fault-free. One can observe that $w = u^i$ and $w^j = v^i$. Therefore, $\langle u, u^i, v^i, v, u\rangle$ be a fault-free 4-cycle. Obviously, (u, u^i) and (v, v^i) is a fault-free edge in Q^0_{n-1} and Q^1_{n-1}, respectively. Since Q^1_{n-1} is fault-free and by Lemma 2, there is an even k-cycle D_k in Q^1_{n-1} which covers the edge (v, v^i) where $k = 4, 8, \ldots, 2^{n-1}$. This implies that a fault-free r-path of $P_r[v, v^i]$ in Q^1_{n-1} exists for $r = 3, \ldots, 2^{n-1} - 1$. Combining edges (u, v), (u, u^i), (u^i, v^i), and the path $P_r[v, v^i]$, the desired even cycle of every length from 6 to $2^{n-1} + 2$ inclusive is obtained.

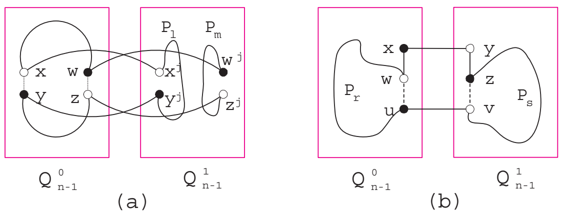

Figure 1: (a). A construction of the desired cycles of lengths from $2^{n-1}+4$ to 2^n inclusive in Subcase 1-3. (b). An illustration of Subcase 2-2.

Subsequently, we will construct a desired even cycle of every length from $2^{n-1}+4$ to 2^n inclusive (see Figure 1 (a)). Since $f^0_e = 2n-6$, let (a, b) be a faulty edge in Q^0_{n-1} such that (a, a^j) and (b, b^j) are two fault-free edges. By induction hypothesis (imagine (a, b) fault-free), (u, v) lies on a fault-free Hamiltonian cycle C of Q^0_{n-1}. One may choose an edge (x, y) in C satisfies that $(x, y) = (a, b)$ if the cycle passes the edge (a, b) or any edge (x, y) in C such that (x, x^j) and (y, y^j) are fault-free if the cycle does not contain the edge (a, b). Consequently, one may choose another edge (w, z) in C such that (w, z) and (x, y) are disjoint as well as (w, w^j) and (z, z^j) are fault-free. Therefore, (w^j, z^j) and (x^j, y^j) are disjoint in Q^1_{n-1}. Since $f^1_e = 0$ and by Lemma 7, two disjoint paths, m-path of $P_m[w^j, z^j]$ and l-path of $P_l[x^j, y^j]$, exist in Q^1_{n-1} where $m = 1, 3, \ldots, 2^{n-2} - 1$ and $l = 1, 3, \ldots, 2^{n-2} - 1$. Merging $P_m[w^j, z^j]$, $P_l[x^j, y^j]$, (w, w^j), (x, x^j), (y, y^j), (z, z^j), and the cycle C, the desired even cycle of every length from $2^{n-1}+4$ to 2^n inclusive is established.

Case 2. $n - 3 \leq f^0_e \leq 2n - 7$ and $f^1_e \leq n - 3$.

Subcase 2-1. $e \in E(Q^0_{n-1}) \cup E(Q^1_{n-1})$. We only discuss the case of that $e \in E(Q^0_{n-1})$ because the discussion of the case of $e \in E(Q^1_{n-1})$ is the same. Since $f^0_e \leq 2n-7$ and by induction hypothesis, the desired even cycle of every length from 6 to 2^{n-1} inclusive is obtained.

Let C be a Hamiltonian cycle of Q^0_{n-1} which contains the edge (u, v). Hence there are 2^{n-2} mutually disjoint edges in C excluding (u, v). Since $2^{n-2} - (2n-5) > 1$ for $n \geq 5$, there exist an edge (x, y) in C such that (x, x^j), (y, y^j), and (x^j, y^j) are fault-free. Hence let $P_{2^{n-1}-1}[x, y]$ be a Hamiltonian path of Q^0_{n-1} which covers the edge (u, v). Since $f^1_e \leq n - 3$ and by Lemma 2, there is a cycle of every even length from 4 to 2^{n-1} inclusive in Q^1_{n-1} which covers (x^j, y^j). This implies that a fault-free k-path of $P_k[x^j, y^j]$ in Q^1_{n-1} exists for $k = 1, 3, \ldots, 2^{n-1} - 1$. Combining $P_{2^{n-1}-1}[x, y]$ and $P_k[x^j, y^j]$ as well as edges (x, x^j) and (y, y^j), the desired even cycle of every length from $2^{n-1}+2$ to 2^n inclusive is established.

Subcase 2-2. e is a crossing edge. Note that $e = (u, v)$. Let $\mathcal{E}(u) = \{i \mid i \neq j \text{ and } (u^i, v^i) \notin F\}$. Suppose that there exists an integer $i \neq j$ such that (u^i, v^i) are fault-free, i.e., $\mathcal{E}(u) \neq \emptyset$. The proof is divided into two parts: **(a)** $N(u) \cap \mathcal{E}(u) \neq \emptyset$ and **(b)** $N(u) \cap \mathcal{E}(u) = \emptyset$.

(a) Suppose that $N(u) \cap \mathcal{E}(u) \neq \emptyset$. Let $i \in N(u) \cap \mathcal{E}(u)$. Hence (u, u^i) and (u^i, v^i) are fault-free edges. By induction hypothesis and Lemma 2 (imagine (v, v^i) fault-free if $(v, v^i) \in F$), there is an even k-cycle C_k in Q^0_{n-1} which covers the edge (u, u^i) where $k = 6, 8, \ldots, 2^{n-1}$; there is an even h-cycle D_h in Q^1_{n-1} which covers the edge (v, v^i) where $k = 4, 8, \ldots, 2^{n-1}$. These imply that a fault-free r-path of $P_r[u, u^i]$ in Q^0_{n-1} exists and a fault-free s-path of $P_s[v, v^i]$ in Q^1_{n-1} exists where $r = 1, 5, \ldots, 2^{n-1} - 1$ and $s = 3, 5, \ldots, 2^{n-1} - 1$. Combining $P_r[u, u^i]$ and $P_s[v, v^i]$ as well as edges (u, v) and (u^i, v^i), the desired even l-cycle is established where $l = r + s + 2$. Therefore, $l = 6, 8, \ldots, 2^n$.

(b) Suppose that $N(u) \cap \mathcal{E}(u) = \emptyset$. Let $i \in \mathcal{E}(u)$. Hence (u^i, v^i) is a fault-free edge and $(u, u^i) \in F$. By the same schemes of (a), the desired even cycle of every length from 10 to 2^n inclusive is obtained. Since $N(u) \cap \mathcal{E}(u) = \emptyset$, $f_c + f_e^0 \geq n - 1$. This implies that $f_e^1 \leq n - 4$. By Lemma 10, let $\langle u, w, x, y, z, v, u \rangle$ be a fault-free 6-cycle where u, w, and x are in Q_{n-1}^0 as well as y, z, and v are in Q_{n-1}^1. Since $f_e^1 \leq n - 4$ and by Lemma 11, a fault-free r-path of $P_r[z, v]$ in Q_{n-1}^1 exists where $P_r[z, v]$ does not pass the vertex y, and $r = 1, 3$. Combining the path $P_r[z, v]$ and the cycle $\langle u, w, x, y, z, v, u \rangle$, the desired 6-cycle and 8-cycle is obtained.

Suppose that every crossing edge $(u^i, v^i) \in F$ for all $0 \leq i \neq j \leq n - 1$, i.e., $\mathcal{E}(u) = \emptyset$. (See Figure 1 (b).) Obviously, $f_c \geq n-1$. Thus, $f_e^0 \leq n-4$ and $f_e^1 \leq n - 4$. Let $\langle u, w, x, y, z, v, u \rangle$ be a fault-free 6-cycle where u, w, and x are in Q_{n-1}^0 as well as y, z, and v are in Q_{n-1}^1. By Lemma 11, (u, w) lies on a fault-free cycle of every even length from 4 to $2^{n-1}-2$ inclusive and each of these cycles does not pass x. Hence a fault-free r-path of $P_r[u, w]$ in Q_{n-1}^0 exists where $P_r[u, w]$ does not pass x and $r = 1, 3, 5, \ldots, 2^{n-1} - 3$. Similarly, a fault-free s-path of $P_s[v, z]$ in Q_{n-1}^1 exists such that $P_s[v, z]$ does not pass y where $s = 1, 3, 5, \ldots, 2^{n-1} - 3$. Combining $P_r[u, w]$, $P_s[v, z]$, the path $\langle w, x, y, z \rangle$, and the edge (u, v), the desired even l-cycle is established where $l = r + s + 4$. Therefore, $l = 6, 8, \ldots, 2^n - 2$. By Lemma 13, the desired Hamiltonian cycle can be found. The proof is complete. □

4 Conclusions

In this paper, we consider the n-dimensional hypercube with some faulty edges such that each vertex is incident with at least two fault-free edges. We use inductively proved scheme to prove that every fault-free edge in Q_n lies on a cycle of every even length from 6 to 2^n provided $n \geq 3$, $f_e \leq 2n - 5$, and every vertex is incident with at least two fault-free edges.

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